

Secondary examination 2019

SCIENCE

Time: 3 1/4 hours

Total marks: 80

General instructions for the examinees.

1. Candidates must write first his or her roll number on the question paper compulsorily.
2. All questions are compulsory.
3. Write the Answer to each question in the given Answer sheet only.
4. For questions having more than one part the Answers to the other part should be written together in continuity.

5. Section	q.no	marks
A	1-11	1
B	12-21	3
C	22-27	4
D	28-30	5

6. There are internal services in question 27 to 30.

SECTION-A

1. Which teeth is used in tearing and chopping of food [1]

Answer: Canines are used for tearing and chopping of the food.

2. Who classified the blood into different groups? [1]

Answer: Karl Landsteiner classified the blood into different groups.

3. Write the full form of CNG. [1]

Answer: Compressed Natural Gas.

4. Write definition of potential energy. [1]

Answer: The energy possessed by a body or a system with respect to its position or arrangement of the particles of the system is called Potential energy.

5. Give one example of renewable resource. [1]

Answer: Solar energy is a renewable energy.

6. Write the scientific name of coffee plant. [1]

Answer: Coffeacaneophora.

7. Define Genetic diversity [1]

Answer: Genetic diversity is the difference in genes among the individuals and species. Genetic diversity gives the living organism unique trait that distinguish them from their closest relatives.

8. Which element takes part in formation of hemoglobin of blood. [1]

Answer: Iron is the element that take part in the formation of hemoglobin.

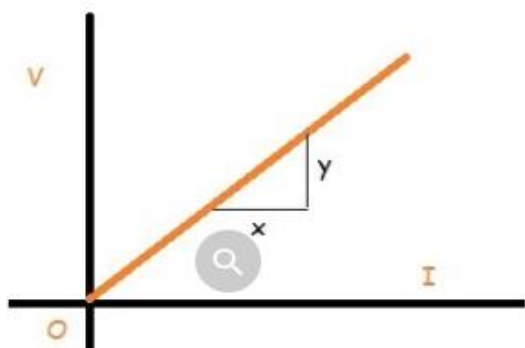
9. Which antibodies participate in allergic reactions?

Answer: IgE isotopes are main antibodies that participate in allergic reactions.

10. Which antigen is found on the surface of red blood corpuscles in addition to A and B. [1]

Answer: Person with blood group A have antigen on the surface of the RBC and person having blood group B have antigen B on the surface of their RBC.

11. Draw a graph between potential difference and electric current according to Ohm's law.



OHM'S LAW

SECTION - B

12. What is the name of natural satellite on the earth? How would have this satellite originated? Mention its two importance for Earth.[1+1+1=3]

Answer: The moon is the only natural satellite of earth and was formed 4.6 billion years ago. Man has moon has been successful in going into the surface of moon and in landing automatic equipment and human being on moon surface and performing their study. The subject of human interest includes promotion of space tourism mining of Helium-3 on the moon and bring it to the earth and using the moon as a Station to travel deep into the space.

13. Explain the process of fossil formation. By which method the age of fossil can be determined. [2+1=3].

Answer: Fossils are the remains of organism which have been preserved in the form of molds and casts in rocks since Prehistoric age's .When a plant or animal dies their remains fall on the ground over a period of time their body gets covered by sediments brought by rivers wind etc. These sediments keep on getting accumulated for hundreds of years and when the land get eroded the Fossil can be seen clearly. The age of fossils can be determined by radiometric dating and relative dating.

14. Write the name of first astronaut. Which facilities are available to solve the problem of food and living in weightlessness for International Space Station? [1+1+1=3]

Answer: Yuri Alekeseyevich Gagarin became the first astronaut. For every Astronaut at International Space station food is sent in plastic bags with the name tag on it. There is a limited facility of keeping the food warm or cold but if it is kept for long time it becomes tasteless. Liquid substance have to the sucked with the help of straw.

As the astronauts lives in weightlessness this at any side effects on their health. To save themselves from this they takes help of exercise .To help exercise in Space Station treadmill and other equipment have been placed.

15) a)Why high beam lights should be used on highways? [1+1+1=3]

Answer: High beam light have illumination so they are used on highways. High beam headlights shine at an angle to illuminate the road twice as far as low beam light.

b) For safe driving which type of mirror is used in headlights and rear view mirror in vehicles?

Answer: Concave mirrors are used in headlights of vehicles as light of the lamp under goes divergence from reflector surface and cover a large area in the front. Rear view mirror of a vehicle are convex mirrors as it gives diminished, virtual and an erect image of the side or rear view.

16)

a) Write the name of pathogen of Naru (Bala) disease.

Answer: *Dracunculus medinensis* is the pathogen of Naru (Bala) disease.

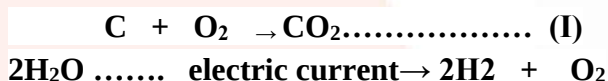
b) Write the name of two alkaloids found in opium.

Answer: Morphine and Narcotine.

c) Write the name of disease caused by chewing Gutka. [1+1+1=3]

Answer: Mouth ulcer and Throat Cancer.

17) In the following chemical equation-



a) Identify the equation 1 and 2 and write its type.

Answer: Chemical reaction (i) is a oxidation reaction called combination reaction. A reaction in which two or more reactants combine to form a single product is called combination reaction. Chemical reaction ii) is a reduction reaction showing electrolysis process. When electric current is passed through water decomposes to give Oxygen and hydrogen.

b) Write any difference between equation 1 and 2.

Answer: In Reaction (i) of Oxidation reaction oxygen is added to carbon to form carbon dioxide while in reduction reaction (ii) oxygen is removed from H_2O .

c) Write one difference between catalytic inhibitor and catalytic promoter.

[1+1+1=3]

Answer: Any inhibitor is a substance capable of stopping or slowing a specific chemical reaction while catalyst is a substance that increases the rate of chemical reaction without being consumed in the process.

18) How does the forest protect fertile soil? Write four measures adopted for the protection of forest. [1+2=3]

Answer: Trees prevent soil erosion by reducing wind, by decreasing over land flow and by increasing infiltration. Soil structure is maintained by reducing desiccation through shading with better soil structure there is a less chance of soil erosion. Four measures for protection of forests are following:

- A massive afforestation programme with emphasis on fuel good and small wood should be launched.
- Efforts should be made to popularize alternative for fuel wood and wood based products.
- Shifting cultivation should be gradually replaced by modern method of farming.
- Measures should be adopted to check and confirm contain forest fires.

19) The ancient Granthacharak Samhita has been written in which language? Explain the knowledge of charak in relation to genetics. [1+2 =3]

Answer: The ancient Grantha charak Samhita Has been written in Sanskrit language.

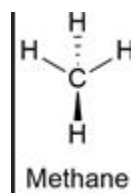
Charak knew the fundamental of genetics. He knew the factors determining the sex of a child. A genetic defect in a childlike lameness or blindness he said is a result of disorder in mother or father which is accepted today.

20) a) Write the name of any one monomer used in terylene.

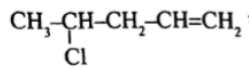
Answer: Monomer used in terylene is Terephthalic acid (1, 4 carboxylic acid).

b) Write structural formula of Marsh gas.

Answer:



c) Write the IUPAC name of
[1+1+1=3]



Answer: 4-chloropentene is the IUPAC name.

21) What are bio medical waste .Mention the name of two diseases caused by them? Explain the process of their disposal by in carnation method. [1+1+1=3]

Answer: Any solid or liquid waste including its container and any intermediate product which is generated during the diagnosis, treatment or immunization of human being, animals, research activities in the production or testing of biological health camps are called by Biomedical Waste.

Diseases caused by biomedical wastes are Meningitis, Hepatitis B, and Hepatitis-C.

SECTION- C

22) a)Which nutritive element is found in abundance in fish.

Answer: Fish is rich source of protein.

b) Give two examples of freshwater fishes.

Answer: katla, Mrigala.

c) What is their diet?

Answer: Fish diet includes water plants small water animals and artificial food such as rice bran wheat scratch pieces of grains etc.

d) How maximum production of fish is done in the pond? [1+1+1+1=4]

Answer: Fish production is dependent on physical, chemical biological quality of water. Environmental factors should be properly managed and regulated for good survival and optimum growth of fish.

23) Explain Mendel's law of dominance with example. Write any two importance of Mendel's law of inheritance [2 +2 =4]

Answer: The law states that when two alternative forms of traits or character are present in an organism only one factor expresses itself in F1 progeny. The factor that expresses itself is called dominant factor while the other factor which remains hidden is called recessive factor. For example when there is a cross between pure tall Plant (TT) and pure dwarf plant only dominant feature expresses itself in the first filial generation. All tall plants were produced masking effect of recessive feature.

Two importance of law of Inheritance:

On the basis of inferences of monohybrid crosses Mendel proposed general laws to consolidate the understanding of inheritance in monohybrid cross. This include law of dominance and law of segregation and Law of assortment. These three laws are called Mendel's law of inheritance. These laws of inheritance is applied to plants and all other organism. They prepare us to make predictions of transfer and inheritance of various Traits, diseases etc. These Laws from the basic of genetics studies

24) a) Give reason for the following statement.

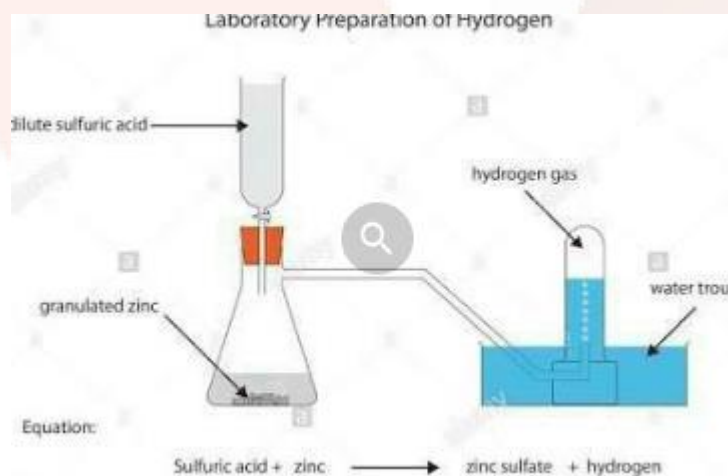
i) Milk of magnesia is used for treatment of acidity in stomach.

Answer: Milk of magnesia is magnesium hydroxide ($\text{Mg}(\text{OH})_2$). It is a base. Stomach acid is a dilute solution of hydrochloric acid a strong acid. They will react chemically to produce magnesium chloride and water It also raises the pH to reduce acidity in the stomach.

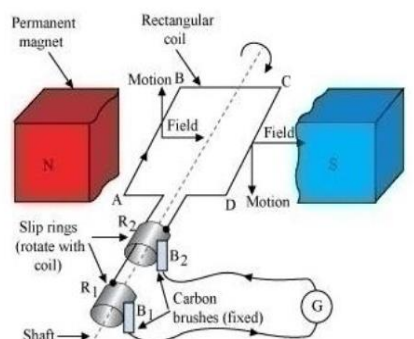
ii) Industrial development of any country is measured on the basis of consumption of sulphuric acid.

Answer: Sulphuric acid has a strong oxidizing and dehydrating properties. It is used in phosphate fertilizers production, mineral processing, metal processing, fibres etc. Increasing demand from various end use industries is considered as one of the primary drivers for sulphuric acid market. Growing production of nutrients dense food crop rising demand for sulphur based fertilizers are the factors that favours industrial development of a country.

b) Draw a labelled diagram of reaction of hydrogen gas. [1+1+2=4]



25) Explain the construction and working of alternating current generator.



AC generator consists of a rectangle armature coil (ABCD) placed in the region of the strong magnetic field between the poles of permanent magnet. It experiences a torque due to the force acting on it and produces a current. Two slip ring maintain contact of coil with the load resistance through brushes without register being removed. As the coil rotates the two rings attached with it also rotate the brushes keep in the contact with rotating ring. When the coil starts rotating with the one arm moving up and other moving down cutting the magnetic lines. According to Fleming's right-hand rule the current is induced in his arms in the direction of ABCD after half the rotation the arm CD moves up and ARM baby moves down as a result the direction of current in each segment changes giving rise to induce current in the direction DCBA.

26)a) A person of 75 kg climbs up to a height of 5 meter in 25 seconds calculate the power consumed by person. ($g = 10\text{m/s}^2$).

Answer:

Mass of person- 75 kg

$G = 10 \text{ m/s}^2$

Distance= 5m

Time- 25 sec

$F = mg$

$= 75 \times 10 = 750\text{N}$

Energy = $F \times ds$

$= 750 \times 5$

$= 3750 \text{ J}$

Power = Energy/ Time

$$= 3750/25$$

$$= 150 \text{ W}$$

b) A block of 9 kg is moving with a velocity of 4 m/s on frictionless surface. The block comes to rest by compressing the spring. If the spring constant is $4 \times 10^4 \text{ N/Kg}$ then find the compression in string.

Answer: $E_p = \frac{1}{2} mv^2$

$$= \frac{1}{2} \times 9 \times 4 \times 4$$

$$= 72$$

Potential energy = $\frac{1}{2} Kx^2$

$$72 \times 2/K = x^2$$

$$144/4 \times 10^4 = x^2$$

$$X = 0.06 \text{ cm or } 6 \text{ m}$$

Q.27) What do you mean by biodiversity? Explain any two importance of biodiversity.

[2+2=4]

Answer: Biodiversity is the existence of a wide variety of plants and animals species living in their natural environment. A rich biodiversity is essential not for ecosystem health. Species from lowest microbes to human plays a part in keeping the planet running smoothly. Each species which is a part is related if a lot of those parts suddenly vanish than the machine that is earth's can't function properly.

OR

What do you mean by Genetic diversity? Explain any two reasons of threat to Biodiversity.

[2 +2=4]

Answer: A single species shows high diversity at genetic level over distributional area. The genetic variation shown by medicinal plant. India has more than five 50,000 different strains of rice and 1000 varieties of mango. This is called genetic diversity. Two reasons to the threat to biodiversity are followings:

1. Habitat loss and fragmentation- This is the most important cause driving animals and plants to extinction

2. Over exploitation- Human have always dependent on nature for food and shelter. It leads to over exploitation of natural resources. Many species extinction the last 500 years were due to over exploitation by human..

SECTION— D

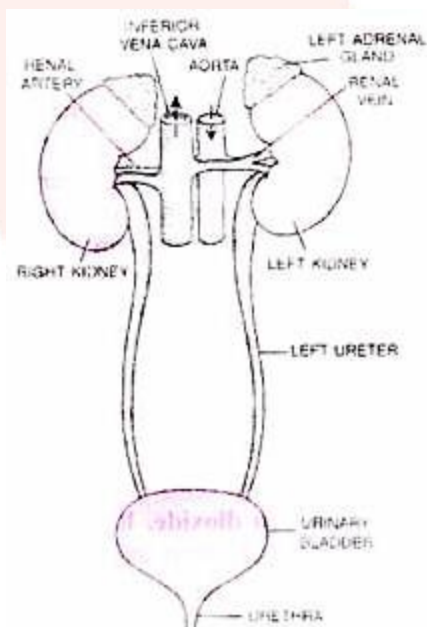
28) a) What is excretion?

Answer: Excretion is a biological process by which an organism removes harmful metabolic waste from the body the mode of excretion is completely different in unicellular and multicellular organisms.

b) Explain the glomerular filtration in urine formation.

Answer: Glomerular filtration is a very step in the process of formation of urine. It involves the filtration of blood into glomerulus. During ultrafiltration the blood flows under a great pressure through Glomerulus. use this high pressure the liquid portion of the blood is filtered out from glomerulus and passes into the Bowman's capsule to enter into the renal tubule this fluid which is entering into the renal tubule is now known as glomerular filtrate for every urine it is hypertonic urine and that is actually accepted out the amount of filtrate formed by kidney per minute is called Normal glomerular filtration rate

c) Draw a well labelled diagram of human excretory system. [1+2+2=5]



EXCRETORY SYSTEM

OR

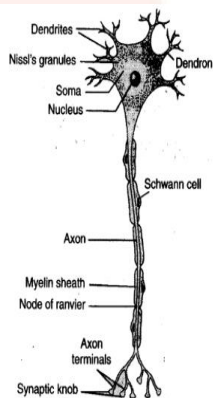
a) Which part of nervous system increases secretion of digestive juices?

Answer: Autonomic nervous system consists of parasympathetic nervous system and sympathetic nervous system. Parasympathetic nervous system increases secretion of digestive tract and sympathetic nervous system decreases the. Parasympathetic nervous system increases the contraction of smooth muscles and increases the rate of peristalsis.

b) Define nerve impulse and synapse.

Answer: A progressive wave of electric and chemical activity along enough fibre that stimulates or inhibits the action of a muscle gland or other nerve cell is called nerve impulse. The small gap between two neurons where nerve impulse passes from one neuron to another in one direction is known as Synapse.

c) Draw a labelled diagram of structure of neuron.



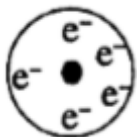
STRUCTURE OF NEURON.

29) a) Explain why atomic size of an element increases from top to bottom in a group?

Answer: The atomic size increases down the group this is because new shells are being added as we go down the group. This increases distance between outermost electrons and nucleolus so that the atomic size increases in spite of increase in nuclear charge.

b) Write the name of scientist who has given the Octave rule for classification of elements. Explain Octave rule.

Answer: John Newlands an English scientist has given the Octave rule for classification of elements. He rearrange the known elements in order of increasing atomic masses and found that every eighth elements had properties similar to that of first. He compare this to the octaves of the music that is sa, re, ga, ma, pa, dha, ni and in the West the user notation do re, mi, fa, so, La ti therefore he called it the lord of octaves. This is known as Newton's law of octaves.



(C)

Write the name of atomic particles present in the centre of represented Rutherford Model. Write one characteristic of this model.

Answer: Nucleus is present in the center.

The characteristics of Rutherford model –

The model describes the atom as a dense positively charged core called a nucleus in which nearly all mass is concentrated around which negative constituent called electrons circulates at some distance like planets revolve around the sun.

OR

a) Explain why metallic character of an element decreases from left to right in a period.

Answer: Elements having a tendency to lose one or more electrons and form positive ions are called metals. As the effective nuclear charge acting on the valence shell electron increases a long period the tendency to lose an electron will decrease so the metallic character of an element decreases from left to right in a period.

b) Write the name of scientist who had given triad Law of classification of elements. Explain the law of triads.

Answer: Johann Wolfgang Dobereiner, a German scientist given triad Law of classification of elements. He has made an attempt to arrange the elements with similar properties into groups having three elements each. He called this group as triads.

c)



Write the name of atomic particle present in the sphere of the represented Thomson atomic model. Explain with an example. [1+2+2=5]

Answer: Protons are present in the sphere. J.J. Thomson proposed that an Atom was a solid sphere of positively charged material and negatively charged particles electrons were embedded in it like this seeds in a watermelon. Edible part of the watermelon is compared with the positive charge in the atom.

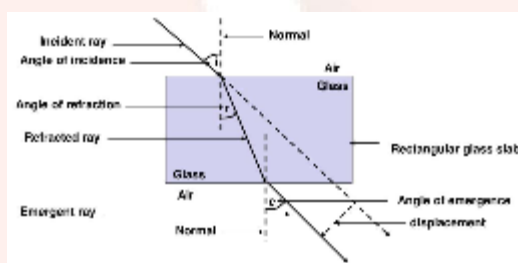
30) a) Write the definition of refractive index of a medium.

Answer: The extent of a change in direction that takes place in in given pair of media is expressed in terms of refractive index. μ_2 represents refractive index of medium 2 with respect to medium 1 when light is travelling from medium 1 to medium 2.

b) What is myopia how this defect can be corrected?

Answer: myopia is a defect in which a person can see nearby objects clearly .but cannot see distant objects clearly. Image is formed before the retina and not on retina. This defect can be corrected by using concave lens of approximate power.

c) Draw a ray diagram of refraction of light in a glass slab .



OR

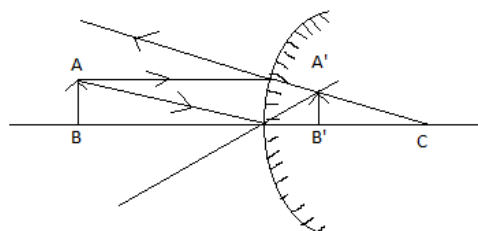
a) What is magnification?

Answer: An increase in the apparent size of an object is called magnification.

b) Define power of accommodation of an eye and the range of eye vision.

Answer: It is the maximum variation in power of Eyes lens for focusing near or for objects clearly at retina. For a young adult with normal vision the power of accommodation is about 4D. The eye loses its power of accommodation at old age.

c) An object is placed between infinity and pole of a convex mirror. Explain the position of image with ray diagram.



The images formed between the principal focus and the pole behind the mirror. The images virtual erects and diminished.

